

TRISTIN WEBB

505-793-0811 | tmwebb@nmsu.edu | linkedin.com/in/tristinwebb | github.com/tristin-w | tristin-w.github.io

EDUCATION

New Mexico State University — Las Cruces, NM

Expected May 2028

Bachelor of Science in Electrical Engineering

GPA: 3.72

- Concentration: AI, Machine Learning, & Data Science

- Minor in Computer Engineering & Software Development

- **Relevant Coursework:** Digital Logic, Computer Organization, Embedded Systems, Data Structures, Signals & Systems, DC/AC Circuit Analysis, Semiconductor & Electronics

San Juan College — Farmington, NM

May 2025

Associate of Science in Computer Science

GPA: 3.96

EXPERIENCE

Undergraduate Research Assistant – WASI Lab, New Mexico State University

Aug 2026 – Present

- Researched and evaluated existing visualization tools and packages for interpreting machine learning models, informing the design approach for an explainable AI (XAI) visualization pipeline.
- Built a directed graph visualization of a reinforcement learning model in Python, applying graph-based data structures to represent state-action relationships and improve model interpretability.
- Studied foundational concepts in reinforcement learning and explainable AI to support visualization design decisions and technical discussions with the research team.

Technology Transfer Intern – Sandia National Laboratories

June 2026 – Aug 2026

- Conducted commercialization and market research for an emerging technology by applying CARAT, Business Model Canvas, Customer Profile, Value Proposition Model, and TAM/SAM/SOM frameworks to assess industry need and adoption potential.
- Expanded industry outreach by sending 200+ LinkedIn connection requests and facilitating 20 customer discovery interviews with professionals across aerospace/defense, industrial manufacturing, and related sectors.
- Identified barriers to entry, customer pain points, and market fit considerations through primary research to inform the technology transfer and commercialization roadmap.

PROJECTS

MIPS Processor Design & Verification | VHDL, Digital Logic, Computer Organization

April 2026

- Designed and implemented a MIPS processor in VHDL, developing core datapath and control logic based on the MIPS instruction set architecture.
- Built a VHDL testbench to verify processor functionality across instruction execution and system-level operations.
- Performed waveform simulation and analyzed timing and signal behavior to debug design issues and validate expected processor operation.

Obstacle-Avoiding Autonomous Car Robot | Arduino, Embedded Systems, C/C++, Arduino IDE

Nov 2025

- Designed and implemented an autonomous obstacle-avoiding robot car using an Arduino microcontroller, ultrasonic sensor, and servo motors.
- Programmed control logic in Arduino IDE to detect obstacles, navigate the environment, and adjust movement in real time.
- Tested, debugged, and demonstrated the final prototype to verify reliable operation and responsive obstacle avoidance.

LEADERSHIP

Officer – Institute of Electrical and Electronics Engineers

Aug 2025 – Present

- Facilitated and supported Open Project Space (OPS) 2.0, coordinating student project activities, collaboration, and event logistics.

Tutor/Mentor – New Mexico State University

Aug 2025 – Present

- Tutored students in Computer Science, Electrical Engineering, Mathematics, Physics, and related subjects.
- Mentored students in developing effective study skills, problem-solving strategies, and academic confidence.

SKILLS

Technical: C, C++, C#, Python, Java, GitHub, VHDL, SQL, Excel, PivotTables, Tableau, MATLAB, Visual Studio, VS Code, Arduino IDE

Analytical: Data Analysis, Data Visualization, Market Research

AWARDS

S-STEM Scholar; Math Excellence Award; Student Leadership Award; Crimson Scholar; Phi Theta Kappa Honor Society; SWASAP Award